

Om Sanjay Singhan

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EDUCATION

New York University, New York, NY— *Dean's List*

September 2023 - May 2025

Master of Science (MS), **Computer Engineering**

GPA: 3.89/4

Coursework: Data Structures & Algorithms, Advanced Machine Learning, Deep Learning, Computational Data Science, Artificial Intelligence, Network Security, Big Data, Computer Systems Architecture, Real-Time Embedded Systems

Mukesh Patel School of Technology Management and Engineering (NMIMS), Mumbai, India

July 2019 - May 2023

Bachelor of Technology in **Computer Engineering**

GPA: 3.76/4

Coursework: Database Management, Machine Learning, Object Oriented Programming, Analysis of Algorithms, Operating Systems

SKILLS

- **Programming Languages:** Python, TypeScript, JavaScript, Java, C#, C/C++, Bash, SQL, R
- **AI & LLM Engineering:** Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), Agentic Workflows, Transformer Models, Model Evaluation, AI Prototyping
- **Backend & APIs:** Node.js, Fastify, Express, GraphQL, REST APIs, PostgreSQL, Prisma, MongoDB, DynamoDB, MySQL
- **Frontend & Data Visualization:** React, Redux, HTML5, CSS, Tailwind CSS, D3.js, Interactive Dashboards
- **ML Frameworks:** PyTorch, TensorFlow, Keras, NumPy, Pandas, NLTK
- **Cloud & Deployment:** AWS, Azure, Docker, Kubernetes, CI/CD, Jenkins

PROFESSIONAL EXPERIENCE

Sewer AI, San Francisco, CA

September 2025 – Present

Software Engineer (*Full-time*)

- **Applied AI System Development:** Built scalable backend services and REST APIs using TypeScript, Node.js, Fastify, and PostgreSQL to support data-intensive, AI-driven inspection workflows serving 10,000+ requests with sub-500ms latency.
- **Test-Driven Development:** Implemented comprehensive test coverage using Jest unit and integration tests and Playwright E2E tests, achieving 85% code coverage and reducing production bugs by 40%.
- **Reliable Distributed Pipelines:** Designed fault-tolerant, state-aware workflows using Temporal.io to orchestrate AI-assisted data processing pipelines handling 10,000+ hours of inspection data daily with 99.5% reliability.
- **Cross-Functional Collaboration:** Partnered with product managers and backend engineers to define metrics contracts and ensure trusted, consistent data across APIs and frontend visualizations.

MAK Capital, New York, NY

February 2025 - May 2025

Artificial Intelligence (AI) Developer (*Intern*)

- **LLM & RAG Systems:** Designed a multimodal Retrieval-Augmented Generation pipeline using GPT-4, LangChain, and Azure Document Intelligence, improving answer accuracy by 40% through semantic chunking and hierarchical retrieval.
- **Prompt Engineering & Evaluation:** Iterated on prompt strategies, vector embedding based retrieval parameters, and evaluation workflows to improve output quality across text and image inputs.
- **Applied ML for Decision Support:** Developed automated NLP pipelines using FinBERT and web-scraped financial data to identify sentiment-driven trading signals, achieving ~80% predictive accuracy.
- **Data Pipelines for AI Systems:** Architected scalable data ingestion, transformation, and storage pipelines using Python, AWS Lambda, and S3, converting raw JSON into governed datasets for analytics, metrics tracking, and exploratory analysis.

Aroris Health, Minneapolis, MN

May 2024 - August 2024

Software Development Engineer (*Intern*)

- **Fullstack API Development:** Improved HIPAA-compliant API infrastructure using Python and Firebase with scalable authentication, protecting 1,000+ patient health records while designing data governance systems for AI-ready pipelines.
- **Product Collaboration:** Collaborated with product managers and UX teams to design and iterate on data-centric frontend features for analytics and reporting workflows.
- **Security-First Design:** Implemented OAuth 2.0 authentication, SQL injection prevention, and encrypted data transmission following OWASP guidelines to protect sensitive patient data.

ACADEMIC PROJECTS

Transforming Distribution System Situational Awareness via Continuous-Time Adaptive Data Fusion - NSF Funded Project

New York University (Generative AI, TimeScaleDB, Express, Python, TensorFlow)

August 2024 - Present

- **AI System Prototyping:** Designed and trained ML models to simulate power-grid behavior, generating 150,000+ synthetic samples with 88% accuracy and implementing continuous feedback loops for model refinement.
- **Time-Series Analytics & Visualization:** Built interactive D3.js dashboards for real-time and historical metrics, improving query performance by 50% while enabling fast experimentation and insights.
- **End-to-End ML Lifecycle:** Owned the pipeline from data ingestion and model training to evaluation and deployment-ready APIs.

Music Recommendation System (Machine Learning, NLP)

July 2022 - May 2023

- **Machine Learning Algorithms:** Implemented collaborative filtering and matrix factorization (ALS, user-user, item-item) to power a personalized recommendation engine.
- **Data Processing & Optimization:** Processed 100k+ song records using Pandas and NumPy, reducing processing time by 40% and improving prediction accuracy by 35%.